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Course: MEE 313
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 300

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Paper 1 of 3 — MEE 313 Manufacturing Technology I Course Handout

Level 300 • First Semester • Handout

LECTURE NOTE

ON

MEE 313

MANUFACTURING TECHNOLOGY I

(METAL CUTTING AND CUTTING TOOLS)

BY

ENGR. PROF. CHIKA E. MGBEMENA

INTRODUCTION

- Metal cutting is a manufacturing process in which a layer of material is removed from a workpiece to achieve the desired shape, size, and surface finish.
- It is a fundamental process in mechanical engineering and manufacturing industries, as it is used to produce components with precise dimensions and tolerances.
- The process involves the use of a cutting tool that shears off material from the workpiece in the form of chips.
- Metal cutting is widely used in operations such as turning, milling, drilling, and grinding.

OBJECTIVES OF METAL CUTTING

- Shape and Size: To produce components with the desired geometry and dimensions.
- Surface Finish: To achieve a smooth surface finish on the workpiece.
- Accuracy: To ensure high precision and tight tolerances in the final product.
- Material Removal: To efficiently remove excess material from the workpiece.

MECHANICS OF METAL CUTTING

The mechanics of metal cutting involve the interaction between the cutting tool and the workpiece. The process can be divided into the following stages:

- Shearing:
 - The cutting tool applies force to the workpiece, causing the material to shear and form chips.
 - The shearing action occurs along a plane called the shear plane.
- Chip Formation:
 - The material removed from the workpiece forms chips, which can be continuous,

discontinuous, or segmented, depending on the cutting conditions and material properties.

- Heat Generation:
- Metal cutting generates significant heat due to friction between the tool and the workpiece, as well as plastic deformation of the material.
- Tool Wear:
- The cutting tool experiences wear over time due to high temperatures, friction, and mechanical stresses.

TYPES OF METAL CUTTING OPERATIONS

- Orthogonal Cutting:
- The cutting edge of the tool is perpendicular to the direction of cutting.
- Commonly used in turning and shaping operations.
- Simplifies the analysis of cutting forces and chip formation.
- Oblique Cutting:
- The cutting edge of the tool is inclined at an angle to the direction of cutting.
- Used in operations like drilling and milling.
- Produces a more complex chip flow and cutting force distribution.

CUTTING TOOL GEOMETRY

The geometry of the cutting tool plays a crucial role in the metal cutting process. Key parameters include:

- Rake Angle:
- The angle between the rake face of the tool and the cutting direction.
- Affects chip flow, cutting forces, and heat generation.
- Clearance Angle:
- The angle between the flank face of the tool and the workpiece surface.
- Prevents the tool
- Cutting Edge Angle:
- The angle formed by the cutting edge with respect to the workpiece.
- Influences the direction of chip flow and cutting forces.
- Nose Radius:
- The radius at the tip of the cutting tool.
- Affects surface finish and tool strength.
- Prevents rubbing against the workpiece.

TYPES OF CHIPS IN METAL CUTTING

- The type of chip formed during metal cutting depends on the material properties, cutting conditions, and tool geometry. Common types of chips include:
- Continuous Chips:
- Formed when cutting ductile materials at high speeds with a sharp tool.
- Provide a good surface finish but may cause tool wear due to heat generation.
- Discontinuous Chips:

- Formed when cutting brittle materials or at low speeds.
- Result in poor surface finish but reduce heat generation.
- Segmented Chips:
- Formed when cutting materials with low thermal conductivity or at high speeds.
- Exhibit characteristics of both continuous and discontinuous chips.

CUTTING FORCES IN METAL CUTTING

• The forces involved in metal cutting can be divided into three components:

- Cutting Force (F_c):
- Acts in the direction of cutting.
- Responsible for material removal.
- Thrust Force (F_t):
- Acts perpendicular to the cutting direction.
- Affects tool deflection and vibration.
- Radial Force (F_r):
- Acts radially outward from the workpiece.
- Influences tool stability and surface finish.

FACTORS AFFECTING METAL CUTTING

• Several factors influence the efficiency and quality of the metal cutting process:

- Material Properties:
- Hardness, toughness, and thermal conductivity of the workpiece material affect chip formation and tool wear.
- Cutting Speed:
- Higher cutting speeds reduce cutting forces but increase heat generation and tool wear.
- Feed Rate:
- The rate at which the tool advances into the workpiece.
- Affects surface finish and material removal rate.

FACTORS AFFECTING METAL CUTTING

- Tool Material:
- The material of the cutting tool (e.g., high-speed steel, carbide, ceramics) determines its wear resistance and cutting performance.
- Coolant/Lubrication:
- Reduces heat generation and friction, improving tool life and surface finish.
- Depth of Cut:
- The thickness of the material removed in a single pass.
- Influences cutting forces and tool life.

TOOL WEAR AND TOOL LIFE

- Types of Tool Wear:

- Flank Wear: Occurs on the flank face of the tool due to friction with the workpiece.
- Crater Wear: Occurs on the rake face of the tool due to chip flow.
- Notch Wear: Occurs at the tool's cutting edge due to high stresses.
- Tool Life:
 - The duration for which a cutting tool can be used before it needs replacement.
 - Governed by the Taylor Tool Life Equation: $V T^n = C$ Where:
 - V = Cutting speed
 - T = Tool life
 - n = Tool life exponent
 - C = Constant

APPLICATIONS OF METAL CUTTING

- Automotive Industry:
 - Manufacturing engine components, gears, and shafts.
- Aerospace Industry:
 - Producing precision parts for aircraft and spacecraft.
- Tool and Die Making:
 - Creating molds, dies, and cutting tools.
- General Manufacturing:
 - Producing machine parts, consumer goods, and industrial equipment.

CLASS WORK

- A high-speed steel (HSS) cutting tool is used in a turning operation on a cylindrical workpiece. The following parameters are given:
 - Cutting speed (V) = 30 m/min
 - Tool life (T) = 90 minutes
 - Taylor's tool life constant (C) = 400
 - Tool life exponent (n) = 0.25

Find:

- The new cutting speed, if the tool life is reduced to 45 minutes.
- How tool life changes if the cutting speed is increased to 40 m/min.

SOLUTION

- New Cutting Speed for Tool Life of 45 Minutes:

According to the Taylor's Tool Life Equation:

$$V T^n = C$$

$$= 400$$

The new cutting speed, is 152.35 m/min.

- Tool Life for Cutting Speed of 40 m/min
- Again using $V T^n = C$
- $= =$
- Hence, The tool life increases to 10,000 minutes when the cutting speed is

set to 40m/min.

Paper 2 of 3 — MEE 313 - MANUFACTURING TECHNOLOGY I

Level 300 • First Semester • Handout

LECTURE NOTE

ON

MEE 313

MANUFACTURING TECHNOLOGY I

(BASIC WORKSHOP MACHINE TOOLS)

BY

ENGR. PROF. CHIKA E. MGBEMENA

INTRODUCTION TO WORKSHOP MACHINE TOOLS

- In manufacturing and fabrication operations, workshop machine tools are crucial pieces of equipment that shape, cut, and finish materials into the necessary shapes.
- These instruments are essential in fields including general engineering, construction, automotive, and aerospace. They make production more accurate, efficient, and repeatable.

CLASSIFICATION OF MACHINE TOOLS

- Cutting Tools:
 - Used to remove material from a workpiece to achieve the desired shape and size.
 - Examples: Lathes, milling machines, drilling machines.
- Forming Tools:
 - Used to deform materials into specific shapes without removing material.
 - Examples: Press machines, forging machines.
- Joining Tools:
 - Used to join two or more components together.
 - Examples: Welding machines, riveting machines.
- Finishing Tools:
 - Used to improve the surface quality of a workpiece.
 - Examples: Grinding machines, polishing machines.

COMMON WORKSHOP MACHINE TOOLS

- Lathe Machine:
 - Purpose: Used for turning, facing, threading, and other cylindrical operations.
 - Components: Headstock, tailstock, carriage, bed, and chuck.
- Milling Machine:
 - Purpose: Used for cutting, drilling, and shaping materials.
 - Types: Horizontal milling machine, vertical milling machine.
- Drilling Machine:
 - Purpose: Used to create holes in materials.
 - Types: Bench drill, radial drill, pillar drill.

- Grinding Machine:
- Purpose: Used for finishing operations to achieve high surface quality and precision.
- Types: Surface grinder, cylindrical grinder, centerless grinder.

COMMON WORKSHOP MACHINE TOOLS

- LATHE MACHINE:
- Purpose: Used for turning, facing, threading, and other cylindrical operations. The main function of a lathe is to remove metal from a piece of work to give it the required shape and size. This is accomplished by holding the work securely and rigidly on the machine and then turning it against a cutting tool which will remove metal from the work in the form of chips.
- Types:
- Speed lathe, Tool room lathe, Engine lathe 6. Special purpose lathe, Bench lathe, NC and CNC lathe, Production lathe (Automatic lathe, capstan and turret lathe)
- Operation: Turning, Facing, Drilling, Boring, Knurling, Taper turning, Thread cutting.

MAJOR COMPONENTS OF THE LATHE

- Bed
- The bed of the lathe forms the base of the machine. It is supported on two legs at a convenient height.
- It carries the head stock and the tail stock for supporting the work and provides a base for the movement of the carriage assembly which carries the tool.
- To ensure accurate machining work it is necessary that the bed has enough rigidly and torsional stiffness to withstand the action of cutting forces.
- The bed of the lathe is sometimes made with a small gap in front of the head stock to accommodate short jobs which need a swing larger than that available on the rest of the bed.

MAJOR COMPONENTS OF THE LATHE

- Head stock
- The head stock houses the spindle and the means for supporting and rotating the spindle.
- It is rigidly fixed on the bed.
- The spindle which is made of steel is made hollow so that long bars which are being machined at the end may pass through it.
- The right hand end of the spindle which projects out of the head stock body has a threaded outside and a tapered bore.

- For turning between centers a carrier plate may be mounted on the threaded end. In larger lathes instead of the threaded end, a flange is provided over which the dog plate, chuck or face plate as the case may be, are located and bolted.
- The tapered end and the hollow spindle also permit mounting of a draw-in collect chuck when designed.

MAJOR COMPONENTS OF THE LATHE

- Tail stock
- The tail stock is for the purpose of primarily giving an outer bearing, support for work being turned on centers.
- It can be adjusted for alignment or non-alignment with respect to the spindle center and carries a center called dead center for supporting one end of the work.
- Both live and dead centers have 60 conical points to fit center holes in the work, the other end tapering to allow for good fitting into the spindles.
- Now-a-days, the dead center is mounted in ball bearing so that it rotates with the job avoiding friction of the job with dead center.
- This is specially necessary with heavy jobs.

MAJOR COMPONENTS OF THE LATHE

- Carriage Assembly
- The carriage assembly of the lathe comprises of a number of components which support, move and control the tool.
- The carriage assembly consists of a saddle, cross slide, compound rest, top slide, tool post and apron.
- Movement of the entire carriage assembly along the bed provides feed for the tool parallel to the lathe axis: movement of the cross slide along its guides on the saddle provides feed of the tool across the lathe axis and the movement of the top slide along its guide over the compound rest provides motion to the tool along a direction set by the compound rest.
- The movement of the carriage and cross slide may be by hand or by power but the movement of top slide is only by hand.

COMMON WORKSHOP MACHINE TOOLS

- MILLING MACHINE :
- Purpose: Used for cutting, drilling, and shaping materials.
- Types: Horizontal milling machine, Vertical milling machine, Column and Knee, Universal Milling Machine, Fixed Bed Milling Machine, Turret Iso called the Bridgeport milling machine, C-frame, Simplex Milling Machine, Duplex Milling Machine, Triplex Milling

Machine, Planner Milling Machine, Tracer

Milling Machine, CNC Milling Machine

- Operations: Plain Milling, Face Milling, Side Milling, End Milling, Gang Milling, Form Milling, Grooves Milling, T-slot Milling, Gear cutting, Thread Milling, Cam Milling, etc.

MILLING MACHINES CONTD.

- A milling machine is a device that rotates a circular tool with symmetrical cutting edges around its axis, holding the workpiece in a vise or similar device.
- These machines are used for machining solid materials like metal, plastic, and wood, and are commonly used to machine irregular and flat surfaces.
- Milling machines are powered tools designed to mill or carve material, using various cutting tools such as rotating blades, drills, and abrasives.
- The best mills feature high-quality cast iron construction, variable speed engines, power feeds, and cutting tools that move along the x- and y-axis.
- They are used in various roles from woodworking to metalworking.
- Milling machines can have self-contained electric drive motors, coolant systems, digital readouts, variable spindle speeds, and power-operated table feeds.

MILLING MACHINES CONTD.

- A milling machine operates by holding a workpiece on a worktable, which is controlled by the table's movement against a rotating cutter.
- The cutter, fixed on a spindle or arbor, moves against the workpiece to remove metal from the surface, producing the desired shape.
- The milling process involves several phases, including milling cutters, surface finish, and gang milling.
- End mills are designed with a special cutting surface for drilling and extended cutting surfaces for peripheral milling.
- They have small cutters at the end corners and are made of highly resistant materials to reduce friction between tools and workpieces.
- Surface finish is achieved through regular intervals of cutting area with regular ridges on the sides, which depend on the feed rate, cutter diameter, and number of cutting surfaces.

MILLING MACHINES CONTD.

- Before starting a cutting job, ensure the workpiece, table, taper in the spindle, and arbor are free from chips, nicks, or burrs.
- Select a milling cutter of the necessary diameter and check the machine's running order, lubrication, and movement.
- Consider the rotation direction, as some cutters can be reversed on the arbor.
- Feed the workpiece in a direction opposite the rotation of the milling cutter (conventional milling).

- Do not change feeds or speeds while the machine is in operation. Secure the workpiece with tight clamps and use recommended cutting oil liberally.
- Use good judgment and common sense in planning every job and set up jobs as close to the milling machine spindle as possible.

APPLICATIONS OF MILLING MACHINES

- Widely used in making gears.
- It is used to produce grooves and slots.
- Milling machines are used to design metal and some other materials.
- Flat and irregular surfaces can be machined on a milling machine.
- Modern milling machines are used to cut super-alloys, titanium, and tensile steel to closer tolerance, greater accuracy, and faster rate of production.

COMPONENTS OF MILLING MACHINES

- Base:
 - This milling machine part is the foundation of milling that carries the weight of the machine and moves it to the ground.
 - All component is mounted on the base.
 - This base is made of cast iron due to its high compressive strength.
- Column:
 - The column is vertically mounted on the base as its supports the knee, table, etc.
 - It also works as housing for all the other driving members.
 - This column is a hollow member that contains driving gears and sometimes motors for the spindle and the spindle and the table.
- Arbor
 - The arbor serves as an extension part of the spindle in a horizontal mill machine.
 - It is fitted to the spindle whenever it's needed.
 - The arbor holds the tool and moves it in the correct direction.

COMPONENTS OF MILLING MACHINES

- Knee
 - The knee supports the saddle and table as its enclosed all-gear mechanism.
 - It is attached to the column in a dovetail way.
 - The knee is designed to be adjusted by a vertical positioning screw (elevating screw).
 - This screw helps to adjust the knee up and down by raising or lowering the lever either by a power feed or by hand.
- Saddle
 - This machine part is located between the table and the knee as it serves as an intermediate part between them.
 - This move transversely to the face of a column which is to provide motion in the horizontal direction to the workpiece.

COMPONENTS OF MILLING MACHINES

- Table
- The table is situated over the knee in order to hold the workpiece during the process.
- It has a t-slot cut on it.
- This provides vertical motion by moving the knee up and down and provides horizontal motion by the feed screw provided by the saddle.
- Overhanging arm
- The milling machine component is used to fasten the arbor support.
- It may consist of one or two cylindrical bars that slide through the holes in the column.
- Spindle
- The spindle is the part of the machine which holds the tool in place.
- It offers rotary motion and drives for arbors, cutters, and attachments used on the machine.

COMPONENTS OF MILLING MACHINES

- Arbor supports
- Milling machines generally have two types of arbor supports, the first one has a small diameter bearing hole, 1-inch in maximum diameter, and the second one has a large diameter bearing hole of about 23/4 inches.
- This milling part contains an oil reservoir that aids the lubrication of the bearing surfaces.
- This arbor support is used only in the horizontal types of milling machines and it's clamped anywhere on the overarm.
- Milling head
- The milling head is the upper part of a vertical mill machine.
- It consists of a spindle, driving motor, and other controlling mechanisms.
- Ram
- The milling machine part can move transversely (in and out) on the column by a hand level.
- One end of the arm is attached to the column and the other ends on the milling head.

COMMON WORKSHOP MACHINE TOOLS

- Drilling Machine:
- Purpose: The drilling process means making a hole in a solid metal piece by using a rotating tool called a drill.
- Types: Sensitive drilling machines, Pillar type drilling machines, Radial drilling machines, Multi-spindle drilling machines.
- Operations: Core drilling, Step drilling, Counter boring, Countersinking, Reaming, Tapping

THE DRILLING OPERATION

- For rotating the drill and to overcome the resistance in cutting, a torque is needed to rotate the drill.
- An axial force is also needed which keeps pushing the drill deeper and deeper into the hole being drilled. The machine feed provides this.
- Machine feed is the downward axial movement of the drill per revolution of the drill.
- The drill will not start cutting the metal if its bottom touches the metal surface superficially.
- This is on account of the chisel edge which does not allow the cutting edges to come into the contact with metal and start cutting until the chisel edge penetrates the metal surface by about an mm or so.
- To facilitate the cutting action, a little depression is made by a punch at the center of the hole to be drilled.

ASSIGNMENT 2

With the aid of a well-labelled hand-drawn diagram, describe the Grinding Machine, highlighting its Purpose, Types and Operations

Paper 3 of 3 — MEE 313 - MANUFACTURING TECHNOLOGY I

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- Consider the rotation direction, as some cutters can be reversed on the arbor.
- Feed the workpiece in a direction opposite the rotation of the milling cutter (conventional milling).

- Do not change feeds or speeds while the machine is in operation. Secure the workpiece with tight clamps and use recommended cutting oil liberally.
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 - All component is mounted on the base.
 - This base is made of cast iron due to its high compressive strength.
- Column:
 - The column is vertically mounted on the base as it supports the knee, table, etc.
 - It also works as housing for all the other driving members.
 - This column is a hollow member that contains driving gears and sometimes motors for the spindle and the spindle and the table.
- Arbor
 - The arbor serves as an extension part of the spindle in a horizontal mill machine.
 - It is fitted to the spindle whenever it's needed.
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COMPONENTS OF MILLING MACHINES

- Knee
 - The knee supports the saddle and table as its enclosed all-gear mechanism.
 - It is attached to the column in a dovetail way.
 - The knee is designed to be adjusted by a vertical positioning screw (elevating screw).
 - This screw helps to adjust the knee up and down by raising or lowering the lever either by a power feed or by hand.
- Saddle
 - This machine part is located between the table and the knee as it serves as an intermediate part between them.
 - This move transversely to the face of a column which is to provide motion in the horizontal direction to the workpiece.

COMPONENTS OF MILLING MACHINES

- Table
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- It has a t-slot cut on it.
- This provides vertical motion by moving the knee up and down and provides horizontal motion by the feed screw provided by the saddle.
- Overhanging arm
- The milling machine component is used to fasten the arbor support.
- It may consist of one or two cylindrical bars that slide through the holes in the column.
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- Types: Sensitive drilling machines, Pillar type drilling machines, Radial drilling machines, Multi-spindle drilling machines.
- Operations: Core drilling, Step drilling, Counter boring, Countersinking, Reaming, Tapping

THE DRILLING OPERATION

- For rotating the drill and to overcome the resistance in cutting, a torque is needed to rotate the drill.
- An axial force is also needed which keeps pushing the drill deeper and deeper into the hole being drilled. The machine feed provides this.
- Machine feed is the downward axial movement of the drill per revolution of the drill.
- The drill will not start cutting the metal if its bottom touches the metal surface superficially.
- This is on account of the chisel edge which does not allow the cutting edges to come into the contact with metal and start cutting until the chisel edge penetrates the metal surface by about an mm or so.
- To facilitate the cutting action, a little depression is made by a punch at the center of the hole to be drilled.

ASSIGNMENT 2

With the aid of a well-labelled hand-drawn diagram, describe the Grinding Machine, highlighting its Purpose, Types and Operations